

Edge Computing Migration

VMware & Cloud to KVM/KubeVirt at the Edge

How enterprises are eliminating VMware licensing at remote sites, repatriating cloud workloads, and deploying lightweight KVM + K3s infrastructure — saving 80%+ on edge compute costs.

hyper2kvm + hypersdk | Proprietary (HyperSDK)

The Edge Challenge

Running VMware at the edge was expensive before Broadcom. Now it's unsustainable.



VMware at Every Site = Multiplied Cost

Each remote site needs vSphere licenses. 50 sites x \$10K/year = \$500K just for hypervisor licensing. Broadcom increases make this 2-5x worse.



Cloud Egress Costs Explode

Edge workloads generate data locally but cloud-hosted VMs pay egress fees. \$0.09/GB adds up to thousands monthly for video, IoT, and analytics workloads.



Latency Kills Cloud Edge

Manufacturing, retail, and healthcare need sub-10ms response times. Cloud round-trip adds 20-100ms. Local KVM eliminates network dependency entirely.



Remote Sites = Limited Resources

Edge locations have constrained hardware, unreliable networking, and no on-site IT. Heavy hypervisors like vSphere waste precious compute on management overhead.

\$500K+

Annual VMware licensing
for 50 edge sites

80%

Cost reduction with
KVM + K3s at edge

10x

Less management
overhead per site

Cloud Repatriation

Why enterprises are bringing workloads back from AWS, Azure, and GCP — and how hyper2kvm makes it painless.

Cloud (Before) • \$8,000-15,000/month per workload • Egress fees on every data transfer • Vendor lock-in (AMI, managed services) • Compliance issues (data residency) • Latency for edge-adjacent workloads • Unpredictable monthly bills	On-Prem KVM (After) • \$500-2,000/month (hardware amortized) • Zero egress fees — data stays local • Open formats (qcow2, libvirt XML) • Full data sovereignty • Sub-millisecond local access • Fixed, predictable costs
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Cost Comparison: 20 VMs Over 3 Years

Cost Item	AWS/Azure	On-Prem KVM	Savings
Compute (VMs)	\$720,000	\$60,000 (hardware)	\$660,000
Storage	\$180,000	\$15,000 (local SSD)	\$165,000
Network / Egress	\$90,000	\$0	\$90,000
Licensing	\$0 (included)	\$0 (KVM is free)	\$0
Migration (one-time)	N/A	\$5,000	-\$5,000
Operations / Support	\$36,000	\$30,000	\$6,000
3-Year Total	\$1,026,000	\$110,000	\$916,000

"We moved 20 VMs from AWS to on-prem KVM and saved \$900K over 3 years. The migration took 2 weeks with hyper2kvm."

Edge Architecture

Lightweight, zero-license infrastructure for every remote site.



Edge Stack (Per Site)

Layer	Technology	Cost
Hardware	Intel NUC / Dell Edge / HPE EL	\$1,000-3,000 one-time
OS	Fedora / RHEL / Ubuntu	\$0 (or \$500/yr RHEL)
Kubernetes	K3s	\$0
VM Runtime	KubeVirt + KVM	\$0
Storage	local-path / Longhorn	\$0
Migration Tool	hyper2kvm	Contact for pricing*
Total per site		\$1,000-3,500

Compare: VMware vSphere Essentials at one edge site = \$6,000-15,000/year in licensing alone. K3s + KubeVirt = \$0/year in software costs.

Industry Use Cases

Edge migration scenarios across sectors.



Manufacturing / Factory Floor

Challenge: SCADA/MES VMs on VMware at 30 plants. \$300K/yr in licenses.

Solution: Migrate to K3s + KubeVirt on ruggedized edge servers. GitOps manages all plants from HQ.

Savings: \$280K/yr. Faster updates, zero license audits.



Retail / Point of Sale

Challenge: POS and inventory VMs in 200 stores on aging ESXi hosts.

Solution: Mini PC with K3s at each store. hyper2kvm converts VMDKs centrally, ship qcow2 images.

Savings: \$1.2M/yr. Standardized deployments, remote management.



Healthcare / Clinics

Challenge: EMR and imaging VMs at 50 clinics. HIPAA requires local data. Cloud egress costs \$40K/mo.

Solution: Repatriate from Azure to local KVM. Data never leaves the clinic. HIPAA compliant by design.

Savings: \$480K/yr in cloud costs. Full data sovereignty.



Energy / Remote Sites

Challenge: Monitoring VMs at 100 substations. Unreliable WAN. VMware licenses at each site.

Solution: K3s on ARM-based edge gateways. VMs run autonomously, sync when connected.

Savings: \$800K/yr. Works offline, self-healing with K8s.



Construction / Temporary Sites

Challenge: Project management VMs deployed per-site, 6-18 month lifecycle. VMware overkill.

Solution: Portable edge box with K3s. Deploy in minutes via USB. Tear down when done. Reuse hardware.

Savings: 90% vs cloud. Zero recurring license cost.



Telecom / Cell Towers

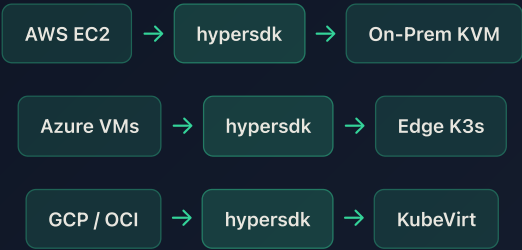
Challenge: vRAN and core network VMs at thousands of sites. NFV licensing spiral.

Solution: Migrate VNFs to KubeVirt on bare-metal K8s. CNF-ready path forward.

Savings: 60-80% on NFV licensing. Path to cloud-native network functions.

Multi-Cloud Exit Strategy

hypersdk supports 10 cloud providers. Export from anywhere, land on KVM.



Cloud Exit Cost Savings

Source	Typical Monthly Cost	On-Prem Equivalent	Monthly Savings
AWS EC2 (m5.2xlarge x 10)	\$5,500	\$800 (amortized)	\$4,700
Azure D8s_v3 x 10	\$4,800	\$800	\$4,000
GCP n2-standard-8 x 10	\$4,200	\$800	\$3,400
S3/Blob storage (10TB)	\$2,300	\$200 (local NVMe)	\$2,100
Data egress (5TB/mo)	\$450	\$0	\$450
Total (10 VMs)	\$17,250/mo	\$2,600/mo	\$14,650/mo

Annual savings for 10 VMs: \$175,800. Payback period on hardware: 2-4 months.

GitOps Edge Management

Manage 100+ edge sites from one Git repo. No SSH, no manual intervention.



Fleet Management

ArgoCD or Flux watches a Git repo. Push a VM manifest change, it deploys to all 50 sites automatically. Rollback by reverting a commit.



Declarative VMs

VMs defined as KubeVirt YAML in Git. Version controlled. Auditable. Reviewable. Same workflow as container deployments.



Zero-Touch Provisioning

New edge site: plug in hardware, boot PXE, K3s auto-joins fleet. ArgoCD deploys VMs. No SSH needed. No on-site IT required.



Central Observability

Prometheus + Grafana federated across all sites. Single dashboard for CPU, memory, disk across entire fleet. Alerting on any anomaly.

Example: Deploy VM to All Edge Sites

```
# git push this to deploy a VM to every edge site:
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: inventory-app
  namespace: production
spec:
  running: true
  template:
    spec:
      domain:
        resources:
          requests:
            memory: 2Gi
            cpu: "2"
        devices:
          disks:
            - name: rootdisk
              disk:
                bus: virtio
      volumes:
        - name: rootdisk
          persistentVolumeClaim:
            claimName: inventory-app-disk
```

Edge Migration Automation

Convert once at HQ, deploy to every edge site automatically.

1. Export from vSphere



2. Convert (hyper2kvm)



3. Push to Registry



4. GitOps Deploy

Centralized Conversion, Distributed Deployment

```
# Step 1: Export + Convert at HQ
sudo h2kvmctl --config edge-vm.yaml

# edge-vm.yaml
cmd: vsphere
vcenter: vcenter.hq.corp
vs_action: export_vm
vs_vm: factory-scada-01
output_dir: ./edge-images
out_format: qcow2
flatten: true
regen_initramfs: true

# Step 2: Upload to container registry as DataVolume source
virtctl image-upload dv factory-scada-disk \
  --size=10Gi \
  --image-path=./edge-images/factory-scada-01.qcow2 \
  --uploadproxy-url=https://cdi-proxy.corp

# Step 3: ArgoCD syncs KubeVirt VM manifest to all edge K3s clusters
# (automatic - just push the YAML to Git)
```



Migration Time per VM

- Export from vSphere: 5-30 min (depends on disk size)
- Convert + fix: 2-10 min
- Upload to registry: 5-15 min
- GitOps deploy to all sites: < 1 min
- **Total: ~15-60 min per VM**



Scale

- Convert 10-50 VMs per day at HQ
- Deploy to 100+ sites simultaneously
- Queue-based processing, no babysitting
- Per-VM boot verification
- **Full fleet migrated in 1-2 weeks**

Total Cost of Ownership

Side-by-side comparison across all approaches.

50 Edge Sites, 5 VMs Each (250 VMs Total), 3-Year TCO

Cost Category	VMware vSphere	Cloud (AWS/Azure)	K3s + KubeVirt
Software Licensing	\$1,500,000	\$0 (included)	\$0
Compute	\$300,000 (hw)	\$4,500,000	\$300,000 (hw)
Storage	\$150,000	\$900,000	\$150,000
Network / Egress	\$50,000	\$540,000	\$50,000
Management Platform	\$200,000 (vCenter)	\$0	\$0 (GitOps)
Operations Staff	\$450,000	\$300,000	\$250,000
Migration (one-time)	N/A	\$50,000	\$30,000
3-Year Total	\$2,650,000	\$6,290,000	\$780,000
Savings vs VMware	—	-\$3,640,000	\$1,870,000
Savings vs Cloud	\$3,640,000	—	\$5,510,000

\$1.87M

Saved vs VMware
over 3 years

\$5.5M

Saved vs Cloud
over 3 years

70%

TCO reduction
vs VMware

88%

TCO reduction
vs Cloud

Next Steps

Start your edge migration journey today.

1

Edge Site Assessment

Inventory VMs at all edge sites. Classify by workload type (always-on, batch, seasonal). Map hardware capabilities. Identify quick wins.

2

Single-Site Pilot

Pick one edge site. Deploy K3s + KubeVirt. Migrate 3-5 VMs with hyper2kvm. Validate: boot, network, application, performance.

3

Fleet Rollout Plan

Based on pilot results, create phased rollout. Batch 1: 5 sites (low risk). Batch 2: 15 sites. Batch 3: remaining. 8-12 week timeline.

4

Decommission Legacy

Cancel VMware licenses site-by-site as migration completes. Repurpose or retire old hardware. Document savings for leadership report.

Save \$1.87M Over 3 Years

Replace VMware at the edge with open source KVM + KubeVirt.
Zero license fees. GitOps management. Production-proven at scale.

hyper2kvm + hypersdk — github.com/ssahani/hyper2kvm

hyper2kvm + hypersdk — Enterprise VM Migration Platform
Proprietary (HyperSDK)